

Problem Analysis and Requirements:

The objective of this project is to build a sentiment analysis system capable of classifying customer product reviews into three sentiment categories: Positive, Negative, and Neutral. Sentiment analysis relies heavily on the quality and consistency of the dataset. A reliable system requires clean, coherent and properly labelled data. Therefore, before the model training, a detailed assessment of data integrity is essential. This aligns with the concept of qualitative error detection, with emphasis on identifying what error to detect, how to detect error, and where to detect error (Chu et al., 2016). The dataset contains 20,000 customer reviews across 8 attributes. There were no missing values; however, during the inspection, some issues were identified that must be resolved too for better modelling performance. First, the dataset is multilingual, containing reviews written in English, French, and Spanish. Since reviews are not in the same language, natural language processing (NLP) pipelines based on a single language struggle to process text correctly, which significantly affects the model's performance. Second, the dataset contains duplicate reviews that are mapped with different star ratings. In supervised learning, identical features mapped to different target labels introduce inconsistencies and degrade predictive reliability (Jordan et al., 2006). Third, although star ratings are provided, the columns are incomplete, as an additional column is needed to map the ratings to the Positive, Negative, and Neutral categories. The training data was split into training and validation during model development. for the purpose of avoiding indirect contamination during hyperparameter tuning and trial of different techniques or processes. Using the test set during model building would violate good machine-learning practice, because it leads to overfitting, where the model learns patterns specific to the evaluation data rather than generalisable behaviour (Müller & Guido, n.d.). This practice ensures that the final reported metrics reflect true generalisation rather than memorisation.

The system must translate non-English reviews into English to maintain linguistic consistency, thereby preventing vocabulary fragmentation and ensuring reliable feature extraction. It must control filter duplicate reviews with different star ratings and resolve label inconsistency. The system must create an additional attribute in the dataset to categorise reviews into positive, negative, and neutral categories, providing a clearly defined target variable for modelling.

Design and Justification:

The development of this sentiment analysis system followed an AI pipeline designed to ensure data quality, linguistic consistency, and robust model performance. After data inspection and visualisation, only three attributes appear relevant for this project: `review_body`, `star`, and `language`. The `review_body` column contains the core textual content used for sentiment classification, serving as the primary feature for model training. The star rating serves as the root label in creating an additional column for the sentiment category. The language column was retained to enable filtering by other languages for translation and to ensure linguistic consistency across the dataset. Selecting only these 3 columns was a result of reducing complexity and preventing unnecessary noise from the non-informative attributes.

- a. **Data Cleaning and Labelling:** Duplicate reviews were addressed to prevent classification imbalance and reduce label noise, both of which can significantly influence model performance. Instead of blindly removing all duplicates, a context-aware deduplication strategy was applied. When duplicate reviews had the same star rating, one instance was retained. For duplicates where the star ratings differed by only one point, the version with the higher rating was kept, as small variations may reflect natural differences in user sentiment. In cases where duplicate reviews had star ratings differing by two or more points, the entries were removed entirely because such inconsistencies represent irreconcilable label conflicts that would introduce harmful noise into the dataset. This method ensured that the data was cleaned in a principled and efficient way, preserving as much useful information as possible while removing elements that could negatively affect model training. To categorise numerical star ratings into meaningful sentiment labels, a mapping function was implemented to map each value to Positive, Neutral, or Negative sentiment. This step was necessary because machine learning models require clearly defined target classes, and raw numerical ratings do not inherently convey the categorical boundaries of sentiment. The function classified low ratings (1–2 stars) as Negative, mid-range ratings (3 stars) as Neutral, and high ratings (4–5 stars) as Positive, reflecting a widely accepted interpretation of customer review scales in sentiment analysis literature. Once defined, this function was applied across the entire dataset to generate a new sentiment column, thereby converting the original numeric ratings into a structured target variable suitable for supervised learning. This transformation ensured consistent label representation and enabled the development of classification models capable of distinguishing among the three sentiment categories.
- b. **Language Handling and Translation Strategy:** Initially, during model training, only English reviews were used, and the model performed well based on F1-score and accuracy. When the dataset was reused with all three languages included, without translation, the model performance degraded significantly. This was due to the text

processing and text representation being designed for English only. After implementing a language-aware TF-IDF representation, the F1-score improved to 41%, which remains relatively poor, likely due to noise and linguistic inconsistency across languages. Instead of moving further with the multilingual dataset or limiting the analysis to only reviews that are English records, which would cause a significant loss of information and introduce avoidable bias, particularly because excluding French and Spanish speakers would contradict ethical principles such as ACM 1.1, which promotes societal well-being, and ACM 1.4, which emphasises fairness and non-discrimination. To ensure that all customer perspectives were represented, translation was adopted as an essential step, especially since French and Spanish are high-resource languages with extensive parallel corpora and share structural similarities with English, which reduces semantic distortion during translation (Wang et al., 2022). Google Translator was selected because it operates as a neural machine translation system capable of producing context-aware translations, and back-translation was implemented to validate translation quality, accompanied by human supervision through random checks to further ensure accuracy. Batch processing was applied during translation to prevent memory overload and maintain efficiency when handling thousands of reviews. Overall, translation improved fairness, preserved diverse customer perspectives, and increased the amount of usable data, which is crucial for achieving stable, accurate classifier performance.

- c. **Classical NLP Methods versus Neural Networks:** Two different text-processing functions were created, one for the classical model and the other for the neural network model. In the classical text-processing function, stop words were removed because they generally contribute little semantic value and tend to inflate noise in machine-learning models; eliminating them reduced computational overhead, but accuracy remained the same across all models before and after removing stop words during training and validation. Negation terms such as “not,” “never,” “cannot,” and “no” were intentionally retained, as removing them can invert sentiment polarity and result in misclassification. for example, converting “not good” into “good”. Stemming and lemmatization were compared to determine the most appropriate normalisation approach for this dataset. Stemming produced higher accuracy, precision, F1-Score and was substantially faster, while lemmatization caused a slight performance drop by 2% on the F1-Score, but the recall was the same, and it was more computationally expensive as it takes time. The factors that determine the selection of stemming over lemmatization were the increase in precision on the Neutral, Positive and Negative, the Recall and the F1 during the training and validation. Even though lemmatization is known for outperforming stemming, in this case, stemming outperformed it. This shows that stemming is more robust for large, informal text such as eCommerce reviews, which often contain slang, abbreviations, spelling variations, and casual phrasing. Contractions were also normalised to ensure that negation meaning was not lost during text cleaning. When punctuation is removed, contractions such as “don’t” or “can’t” can be misinterpreted as fragmented forms like

“don t” or “can t,” which lack semantic value and prevent the model from recognising negation. To avoid this, contractions were expanded prior to noise removal, converting “don’t” to “do not” and “can’t” to “cannot,” which preserves the polarity structure of the original message. Neural Network text processing is different from the classical machine learning because the learning methods of these models are different as neural network learns semantic and syntactic patterns directly from sequence of words, Neural network learns with context cue, word form which are essential for learning and capturing sentiment expressed through phrasing rather than just individual words. The preprocessing function steps for the neural network are only lowering casing, contraction expansion, and removing non-alphabetic characters.

- d. **Feature Extraction and Classical Model Selection:** Two feature extraction methods Bag-of-Words (BOW) and Term Frequency Inverse Document Frequency (TF-IDF), were evaluated alongside three classifiers: Logistic Regression, Naïve Bayes, and Support Vector Machines (SVM). On the Bag of Word, all models appear to perform well, when looking at the accuracy of alone (about 66%-69%), showing that accuracy alone is not enough to choose the best model. I noticed that Support vector machine on Bag of Word struggles to correctly identify Neutral reviews because of the mixed and subtle opinions on the neutral sentiment and the Bag-of-Words only counts words and ignores context, so it cannot capture these subtleties well, As a result, the Neutral class has a low F1-score, even though overall accuracy seems acceptable. Naïve Bayes with Bag-of-Words achieves the highest overall accuracy among the three models, but this improvement is mainly due to stronger recall for the Positive class (0.81) and does not result in better performance on the Neutral class, which remains weak with an F1-score of 0.36. Logistic Regression with Bag-of-Words shows more balanced and stable performance across the majority classes and avoids extreme class bias; however, its recall score for Neutral sentiment is (0.42) which is the highest recall score when training and validating and the F1 Score is 66% which is the lowest among the 3 models. Given the linguistic ambiguity of Neutral sentiment, this suggests that model selection should be guided by class-wise performance rather than overall accuracy alone. For the TF-IDF-based classifiers, all three models achieve a similar overall accuracy of approximately 69–70% on the validation set. When performance was analysed based on classification behaviour, Logistic Regression showed balanced performance on the Negative and Positive classes and demonstrated the strongest capability in handling Neutral sentiment, achieving a recall of 0.47 and an F1-score of 0.42. This shows that Logistic Regression is more sensitive to linguistically ambiguous reviews, though it yields slightly lower overall accuracy than SVM. Multinomial Naïve Bayes improved raw accuracy and exhibited very strong recall for the Negative (84%) and Positive (85%) classes; however, this was achieved at the expense of Neutral sentiment detection, where recall dropped to 7%. This indicates a strong bias towards polar sentiment classes, effectively collapsing Neutral instances into Negative or Positive predictions. Consequently, Naïve Bayes is unsuitable

for a three-class sentiment classification task where Neutral detection is important. SVM with TF-IDF demonstrates the most consistent and robust trade-off across all sentiment classes, achieving the highest overall accuracy (70%), the highest weighted F1-score (69%), and a well-balanced macro-average F1-score (64%), indicating stronger generalisation across classes. Although the Neutral recall (36%) is lower than that of Logistic Regression, SVM shows a more balanced overall performance. This leads to the conclusion that SVM benefits most from TF-IDF representations, effectively capturing discriminative features while maintaining stability across sentiment categories. In conclusion, SVM with TF-IDF demonstrated the most balanced and robust performance during validation and tuning, particularly for Neutral sentiment detection.

- e. **Neural Network Design and Justification:** CNN and BiLSTM architectures were explored on this project. Hyperparameter tuning revealed that increasing model capacity often improved performance for dominant classes but risked overfitting and reduced Neutral recall. For CNN, increasing the output from 128 to 200, the overall F1-score remained unchanged, but at the classification level, the F1-scores for the Negative and Positive classes increased by 1%, while the Neutral class dropped by 1%. When the output was increased to 300, the overall F1-score dropped by 2%. Although the `kernel_size` was tuned from 3 to 5 to capture different sentiment cues, the F1-score, as well as the classification Precision and Recall, decreased by 3%, making it necessary to return the `kernel_size` to 3. The number of training epochs was then increased from 5 to 10, but this did not result in a significant change in performance. However, increasing the epoch to 20 improved the Neutral class Recall from 29% to 33%, representing a modest but acceptable improvement. Even after increasing the epoch to 20, the overall F1-score dropped by 2%, which is not an issue because I am really concerned about the Recall scores at the classification level. The batching size helps prevent memory from crashing. The BiLSTM model was observed to be slow and computationally expensive when configured as Bidirectional (LSTM(128)). Reducing the number of units to 64 significantly improved training speed. Additionally, increasing the batch size to 64 further accelerated computation without negatively affecting model performance. When the Dense layer was initially set to Dense(64) and later increased to Dense(120), the recall of the Neutral class improved by 2%, indicating a performance gain. However, the Dense layer size was subsequently reduced back to 64 to mitigate overfitting, as excessively large Dense layers can disproportionately increase model capacity without introducing additional informative features.

BiLSTM and CNN models were explored due to their ability to learn distributed representations and capture sequential or local patterns in text (Xiong et al., 2024) from the validation result, the CNN achieved higher overall validation accuracy and stronger weighted F1-score than the BiLSTM, the CNN showed improved recognition of negative and positive sentiments which dominate the dataset, while maintaining comparable

macro-level performance. Prior studies have shown that CNN-based models often match or outperform recurrent architectures for sentiment analysis while converging faster and exhibiting improved training stability (Johnson & Zhang, 2017). Therefore, considering both the observed experimental outcomes and the linguistic structure of the dataset, CNN was chosen as the preferred neural network model for customer review sentiment analysis.

Implementation

The dataset was loaded, and an initial inspection of dataset size, schema, language distribution, and star-rating frequencies was conducted to identify imbalance and data quality issues. A context-aware duplicate-removal strategy was then applied, resolving identical reviews based on the consistency of their star ratings to reduce label noise. To resolve multilingual review issues, the non-English review was translated into English using a Google Translate pipeline, then back-translation. Human validation was performed on some translated reviews to assess whether they matched the translation and back-translation. Sentiment labels were derived from star ratings using a rule-based mapping into positive, neutral, and negative classes. The training data was split into training and validation sets using a stratified 80/20 split to preserve class proportions. Text preprocessing was performed using two tailored pipelines: a classical model pipeline that included contraction expansion, noise removal, stop-word filtering with negation preservation, and stemming, and a lighter preprocessing pipeline for neural models that preserved word order. Feature extraction for classical models employed Bag-of-Words and TF-IDF representations with controlled vocabulary sizes, while neural models used tokenised and padded word sequences. Sentiment labels were encoded numerically and converted to a categorical format for neural networks. Feature extraction for classical models employed Bag-of-Words and TF-IDF representations with controlled vocabulary sizes, while neural models used tokenised and padded word sequences. Sentiment labels were encoded numerically and converted to a categorical format for neural networks. The same rule-based preprocessing pipeline was applied to the training, validation, and test datasets to ensure consistency in text representation. Since these preprocessing steps are deterministic and do not involve parameter learning, they do not introduce data leakage. All learned components, such as the TF-IDF vectorizer, were fitted exclusively on the training data and subsequently applied to the validation and test sets.

Evaluation and Results

The evaluation was conducted on the actual test data consisting of 1,500 unseen reviews, using accuracy, precision, recall, and F1-score as performance metrics. Considering the linguistic ambiguity associated with Neutral reviews, macro-averaged metrics and class-wise recall were prioritised over raw accuracy, as accuracy alone is known to be misleading under class imbalance and asymmetric error costs.

The Bag-of-Words (BoW)-based models achieved moderate test accuracy, with Naive Bayes attaining the highest accuracy (0.67). On the test set, it showed improved Neutral recall (0.33), indicating better performance on unseen data compared to the training and validation phases. Logistic Regression with BoW exhibited more stable class-wise behaviour than Naive Bayes, achieving the highest Neutral recall among the classical BoW models (0.43). However, this improvement came at the cost of reduced precision and an overall accuracy of 0.64. This trade-off occurred because the model became better at identifying Neutral instances while increasingly misclassifying other sentiments as Neutral, a compromise reflected in the macro F1-score. The SVM with TF-IDF representation achieved an accuracy of 0.653 and demonstrated relatively balanced performance across the Negative and Positive classes. However, its Neutral recall remained limited (0.30). Although TF-IDF provides improved term weighting by emphasising informative words and down-weighting common ones, the linear, margin-based SVM still struggled to distinguish weakly polarised sentiment, particularly around the linguistically ambiguous Neutral class. The F1-score of 0.65 indicates consistent performance across most classes but also highlights the model's difficulty with Neutral reviews. Overall, classical models benefited from TF-IDF representations compared to BoW; however, none achieved strong generalisation on the Neutral class. This highlights the representational limitations of frequency-based features for capturing ambiguous sentiment.

The BiLSTM achieved a test accuracy of 0.66, outperforming most classical models except Naïve Bayes. It achieved improved recall for the Negative class (0.80), indicating a strong ability to capture sentiment expressed through longer contextual dependencies. However, Neutral recall remained low (0.31), showing that even as a sequential model, the BiLSTM does not fully resolve the semantic ambiguity inherent in Neutral reviews. The CNN model achieved a comparable accuracy of 0.659 and showed the strongest recall for the Negative class (0.86). However, it performed worst on the Neutral class, with a recall of only 0.22. This outcome reflects the architectural bias of CNNs toward detecting strong local features, which are less prevalent in linguistically balanced or weakly opinionated Neutral reviews.

Across both neural models, weighted F1-scores were comparable to those of the classical models, and macro-averaged F1-scores did not show significant improvement. This indicates that increased model complexity did not translate into proportional gains in class-balanced performance, particularly for Neutral sentiment detection. Overall, the results demonstrate that accuracy alone is insufficient as an indicator of model effectiveness for this task. Models such as

Naive Bayes achieved higher accuracy by prioritising dominant sentiment classes, while Logistic Regression and BiLSTM offered slightly improved sensitivity to Neutral sentiment at the expense of precision. Neural models excelled in recognising strongly polarised sentiment but failed to meaningfully improve Neutral classification, suggesting that Neutral sentiment ambiguity is driven more by data characteristics than model capacity. These findings confirm that class-wise evaluation is essential for sentiment analysis, particularly when Neutral sentiment plays a critical role. The results also suggest diminishing returns from increased architectural complexity without targeted strategies such as class rebalancing, ordinal sentiment modelling, or the incorporation of auxiliary semantic features.

Conclusion and Limitations:

This project focused on customer review sentiment analysis using an end-to-end pipeline, covering data loading, inspection, deduplication, translation from other languages into English, sentiment labelling from star ratings, text processing, feature extraction, data splitting, and model evaluation. Both classical machine learning models (Logistic Regression, Naïve Bayes, and Support Vector Machines with Bag-of-Words and TF-IDF features) and neural network models (BiLSTM and CNN) were implemented and assessed. The outcomes show that TF-IDF-based classical models consistently outperform Bag-of-Words representations, confirming that weighting terms by importance rather than raw frequency is more effective for sentiment classification. Among the classical approaches, SVM and Logistic Regression with TF-IDF provided the most balanced performance across sentiment classes. Neural models achieved competitive results but did not fully outperform classical methods.

Limitations to this project include: first, sentiment labels were derived from star ratings, which might not always align perfectly with the textual sentiment expressed in the review, introducing label noise. Second, machine translation may distort sentiment cues, especially for idiomatic expressions. Third, the Neutral class remains difficult for all models due to its small size, as other classes have much more data than it does. These limitations could be addressed by experimenting with advanced multilingual or pre-trained language models, such as BERT, and by using an equal or balanced dataset for the three classifications.

Link to the translated files

https://drive.google.com/drive/folders/1QzOYZD--xVtjN19-l5nvJ8Z4PRQ0Rnp3?usp=drive_link

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